

Supplementary Information: Active SolvAI v3 adaptive simulation effort

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1 Prospective scope and provenance

The authoritative blueprint has SHA-256 264f7f4afad7a9fa578992b33e6d5fef83fe15f74be7fb9d85fd7e2f751b4de. The inherited evidence audit and v3 master protocol were committed before model fitting. The branch descends from the v2 power-gate commit 770e13ce68ca80104557fd67a224abc5a2c44767. The v1 and v2 no-go commits are 8fb984c2eb26d016c6b81cf488f88dc667ca9cd3 and f2cc60fb73416e0e417c97c600d5abd00868031c, respectively.

The parent ARROW set, Tier-A, 76 historical PIMD2-probe molecules and all twelve full-grid molecules are development-exposed. Exact and standardized-equivalent identities cannot enter a future sentinel. Experimental hydration labels were not read by the v3 analysis code.

2 Inherited trajectory inventory

The workspace search found 418 ‘SYSTEM’ energy files. Of these, 363 were compatible qualified PIMD2 outputs: 219 successful historical probe outputs and 144 new dense-grid windows. Seventy-two molecules had complete 0.1/0.5/0.9 observations. The dense library comprised twelve molecules and fifteen windows; 36 windows reused historical outputs and 144 were generated in v1. Every molecule-window had one 5-ps stream and no compatible independent replica.

Table 1: Dense development molecules. All are development-exposed.

Calibration	Prospective in v1
Propane	Cyclohexane
Ethanol	Octane
Acetone	Octanol
Pyridine	AceticAcid
	Acetamide
	DiMethylEther
	EthylAcetate
	Anthracene

3 Sequential replay implementation

The replay library stores a rectangular window-by-chunk-by-observation tensor but exposes a policy only to a frozen state containing lambda, revealed chunk counts, revealed means, revealed SEMs, cumulative cost and step number. A policy cannot access the raw chunk tensor or future reference. Tests verify deterministic equal-time and random replay, recovery of a visibly high-variance synthetic window by

max-SEM and rejection of an ineligible future action. The inherited one-stream library is used for implementation tests only.

4 Gate-1 feature sets

Table 2: Predeclared model ladder.

Model	Inputs
Lambda/protocol	lambda, lambda squared, endpoint indicator, trapezoidal weight
Structure cold start	lambda/protocol; molecular weight, heavy atoms, TPSA, MolLogP, H-bond donors/acceptors, rotatable bonds, rings, fraction Csp3, formal charge; fixed lambda interactions
SolvAI cold start	lambda/protocol; 15 frozen response coordinates; fixed lambda interactions
Generic observed	lambda/protocol plus prefix mean/variance/SEM, batch and spectral variance rates, autocorrelation, effective sample size, half-prefix distances, drift, temperature, density and equilibration flag
Structure-conditioned	generic observed plus ordinary structure features and fixed lambda interactions
SolvAI-conditioned	generic observed plus the 15 response coordinates and fixed lambda interactions

5 Complete fixed-prefix results

Table 3: Primary stabilized-target results across every fixed prefix. Lower MAE is better.

Prefix (ps)	Lambda/protocol	Generic observed	Structure-conditioned	SolvAI-conditioned
0.5	1.547	1.438	1.485	1.466
1.0	1.547	1.427	1.488	1.475
2.0	1.547	1.430	1.431	1.438
3.0	1.547	1.444	1.423	1.460

At the primary prefix, SolvAI minus generic was +0.0477 (90% confidence interval, -0.0171 to +0.1199). Aligned SolvAI minus the mean shuffled control was -0.0102 (90% confidence interval, -0.0478 to +0.0310). Sensitivity stabilizers of 0.10 and 0.50 kcal mol⁻¹ gave SolvAI-minus-generic effects of +0.0620 and +0.0254.

6 Power and reference reliability

The minimum-useful effects were -0.341905 and -0.222900 kcal mol⁻¹ at five and seven windows. The reference threshold was 0.111450 kcal mol⁻¹. Fifty thousand Monte Carlo cohorts were generated for each predeclared molecule count and duration.

The reference calculation assumes inverse-square-root noise decay and is therefore optimistic if slow relaxation or initialization bias persists. The effect calculation uses v1 policy heterogeneity because no independent v3 policy effect exists. These limitations make the design a lower-bound planning diagnostic, not a launch proposal.

Table 4: Selected power/reference designs. The final row is the first adequate point in the predeclared grid.

N	ps/stream-window	Power ₅	Power ₇	Reference pass	Reserved GPU h
12	50	0.218	0.114	0.000	54.0
20	50	0.326	0.142	0.000	90.0
20	3000	0.326	0.142	0.734	5401.4
64	3000	0.769	0.317	0.716	17284.6
128	3000	0.970	0.545	0.821	34569.2
256	3000	1.000	0.830	0.827	69138.3

7 Negative-result disposition

No policy model was selected and no prospective cohort was qualified. No new trajectory, experimental-endpoint analysis, PIMD4/PIMD8 calculation or Tier-B test was opened. The v3 result is a causal stop: the frozen response coordinates failed to add sampling-difficulty information, and the minimum powered independent design was not compatible with the local exploratory envelope.

8 Reproduction and artifacts

The exact commands are listed in the workstream README and Makefile. Machine-readable predictions, bootstrap contrasts, diagnostic reliability, the complete power grid, file hashes and the append-only run ledger are retained under `active.solvai_v3/results`, `data` and `runs`.